

Question:

What is the de Broglie wavelength of an electron with a kineticenergy of 1eV?

Answer Choices:

- A. 2.48×10^{-8} cm
- B. 5.92×10^7 cm/sec
- C. 8.66×10^{-6} cm
- D. 3.14×10^{-5} cm
- E. 9.81×10^{-4} cm
- F. 5.27×10^{-9} m
- G. 1.23×10^{-7} cm
- H. 4.14×10^{-10} m
- I. 6.625×10^{-27} erg sec
- J. 0.512×10^6 eV

Reasoning + Answer:

<think>
To find the de Broglie wavelength of an electron with a kinetic energy of 1 e V , we need to follow these steps :

- 1 . Convert the kinetic energy from e V to Jou les : $1 \text{ e V} = 1.602 \times 10^{-19} \text{ J}$, so $1 \text{ e V} = 1.602 \times 10^{-19} \text{ J}$.
- 2 . Use the kinetic energy formula to find the momentum : $\text{kinetic energy} = 0.5 \times \text{momentum}^2$, rearrange to get $\text{momentum} = \sqrt{2 \times \text{kinetic energy}}$
- 3 . Convert the momentum from kg*m/s to J*s , this factor having no value in momentum equation works by De Broglie relation .
- 4 . Use the de Broglie relation : $\lambda = h / p$, where h is the Planck constant and p is the momentum .
- 5 . calculate free length In all formula constants

Planck's constant = $6.626 \times 10^{-34} \text{ kg} \cdot \text{m}^2 \cdot \text{s}^{-1}$
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 $5.27 \times 10^{-9} \text{ m}$